

Test Report - VxMark Cold and Hot Storage of Goldenmate 1500VA/1000W UPS and HP4001dn Printer

Author: Pius Wong , created Apr 30, 2026 , updated Jun 3, 2026

Summary

The Goldenmate 1500VA/1000W uninterruptible power supply (UPS) and HP4001dn laser printer are likely to pass standard low- and high-temperature storage tests defined in VVSG 2.7-F at -4°F (-20°C) and 140°F (60°C). This UPS and printer are intended for use with the VxMark ballot-marking device (BMD). The Goldenmate UPS retained normal function in the VxMark system after 4-hour soaks at cold and hot temperatures, unpackaged. The printer retained normal function in the VxMark system after a 4-hour soak at the hot temperature, unpackaged, if the standard HP148A toner is replaced after hot storage. The printer is already rated for cold storage at -4°F by the manufacturer.

Applicable VVSG Requirements

- 2.7-D – Ability to support maintenance and repair physical environment conditions – non-operating
- 2.7-E – Ability to support transport and storage physical environment conditions – non-operating
- 2.7-F – Ability to support storage temperatures in physical environment – non-operating

Devices Under Test

- Goldenmate 1500VA/1000W UPS
- HP4001dn printer

Purpose

The VxMark system must be able to withstand storage in extreme cold and hot temperatures, from -4°F (-20°C) to 140°F (60°C), and from 25% relative humidity (%RH) to 55%RH, according to VVSG 2.7-F. This requirement also informs other VVSG requirements 2.7-D and 2.7-E listed

above. This means the VxMark BMD, its backup UPS, and its printer all must withstand these conditions.

The VxMark BMD itself has no parts or subassemblies that are expected to fail in this environmental range, based on manufacturer rated storage specifications. Furthermore, VxMark shares many critical electronic and hardware components with VxScan, and past extreme environmental storage tests on VxScan showed passing results both in internal and external testing.

However, storage test data on the Goldenmate 1500VA/1000W UPS and the HP4001dn printer were limited. Standard 4-hour cold and hot storage tests were planned for the UPS, mimicking past tests performed on VxSuite products at Element in Longmont, CO. The hot storage test was planned for the printer to run simultaneously with the UPS. The HP printer is already explicitly rated for storage in the cold conditions by the manufacturer and was left out of the cold test.

Materials

- Goldenmate 1500VA/1000W UPS (JUO2HE390707)
- HP4001dn laser printer (VNL0728678), with:
 - compatible 28lb bond paper
 - HP148A toner
- VxMark unit (MK-14-018) for system testing
 - Imaged with v4.0.4 production app
 - Configured with summary ballot test election
- VxScan unit for verifying printed ballots
- Temperature Chamber (Thermotron S-16-8200 administered by Austin Reliability Labs in Austin, TX)

Procedures

Specific steps were as follows:

1. Check the UPS and printer for normal function before storage tests.
 - a. Follow tests of normal function (*VxMark Tests of Normal Function*) to confirm the test unit functions normally before any disruptive tests.
 - b. Document tests of normal function. Record any pre-existing problems. Use the pre-existing template.

- c. Confirm the UPS is at a state of charge between 40-100%, and turn it off.

2. Store the UPS in cold conditions for 4-hour dwell.

- a. Place the UPS in the temperature chamber, unpackaged.
- b. Configure the test chamber as follows:
 - i. -4°F (-20°C), 4-hour dwell
 - ii. Humidity uncontrolled due to low temperature
 - iii. <3°C/min ramp-up and ramp-down
 - iv. Chamber sits at ambient for several hours before opening to retrieve sample.

3. Check the UPS for normal function after cold storage.

- a. Follow tests of normal function to confirm the test unit still functions normally.
- b. Document tests of normal function. Note if any functions have changed.

4. Store the UPS and printer in hot conditions for 4-hour dwell.

- a. Place the UPS in the temperature chamber, unpackaged.
- b. Configure the test chamber as follows:
 - i. -140°F (60°C), 4-hour dwell
 - ii. Humidity at 55%RH
 - iii. <3°C/min ramp-up and ramp-down
 - iv. Chamber sits at ambient for several hours before opening to retrieve sample.

5. Check the UPS and printer for normal function after hot storage.

- a. Follow tests of normal function to confirm the test unit still functions normally.
- b. Document tests of normal function. Note if any functions have changed.

Results

1. Normal function was confirmed for the UPS and printer before storage tests.

- a. See photos of the units before testing in [Appendix A](#) and [Appendix B](#), taken by Austin Reliability Labs on-site.
- b. The UPS and printer were tested with the VxMark BMD. All essential functions worked as expected, documented in spreadsheet [Appendix C](#).

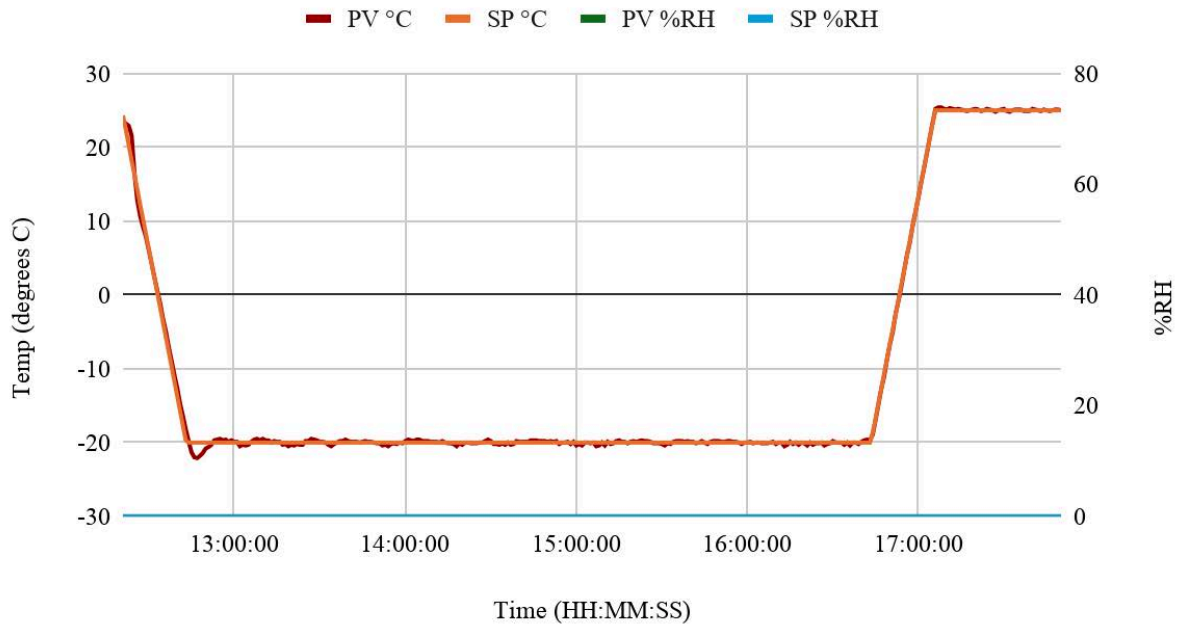
2. The UPS was stored in cold conditions for a 4-hour dwell.

- a. The test setup inside the chamber is shown below, before closing the chamber door:



- b. Test conditions followed the intended set point (SP) for the process variable (PV) of temperature, with humidity at 0%RH uncontrolled. This is shown in the graph below of the chamber data log:

Temperature & %RH vs. Time, Cold Storage



3. Normal function was confirmed for the UPS after cold storage.

- See photos of the UPS after cold storage in [Appendix A](#).
- After resting overnight, the UPS and printer were tested with the VxMark BMD. All essential functions worked as expected, documented in spreadsheet [Appendix D](#).

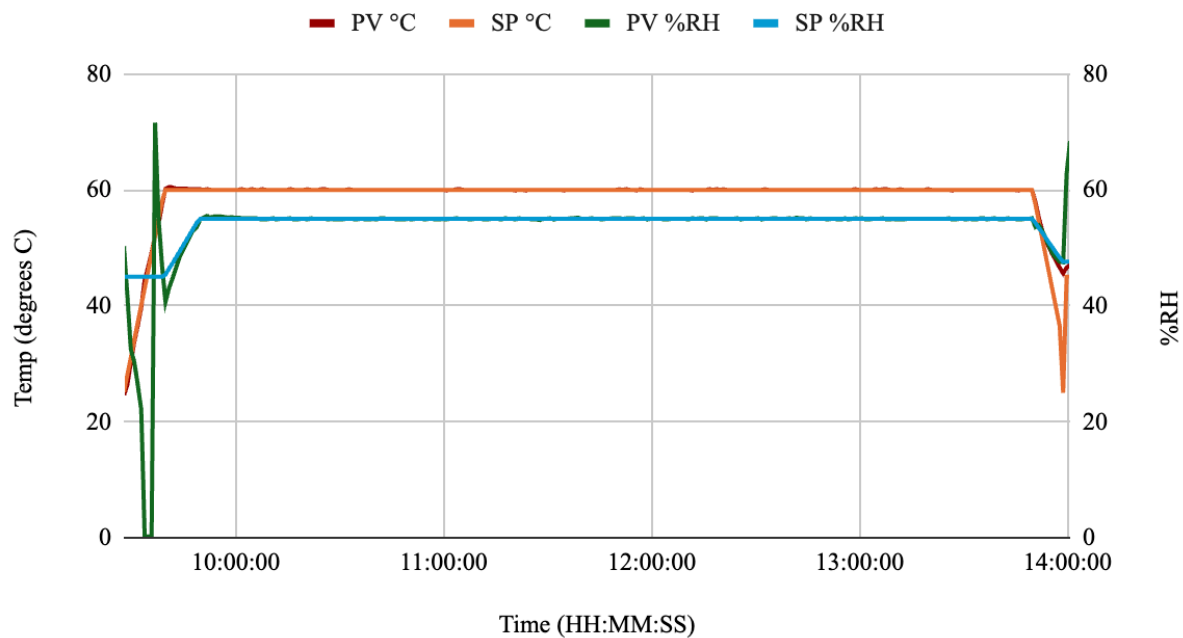
4. The UPS and printer were stored in hot conditions for a 4-hour dwell.

- The test setup inside the chamber is shown below, before closing the chamber door:



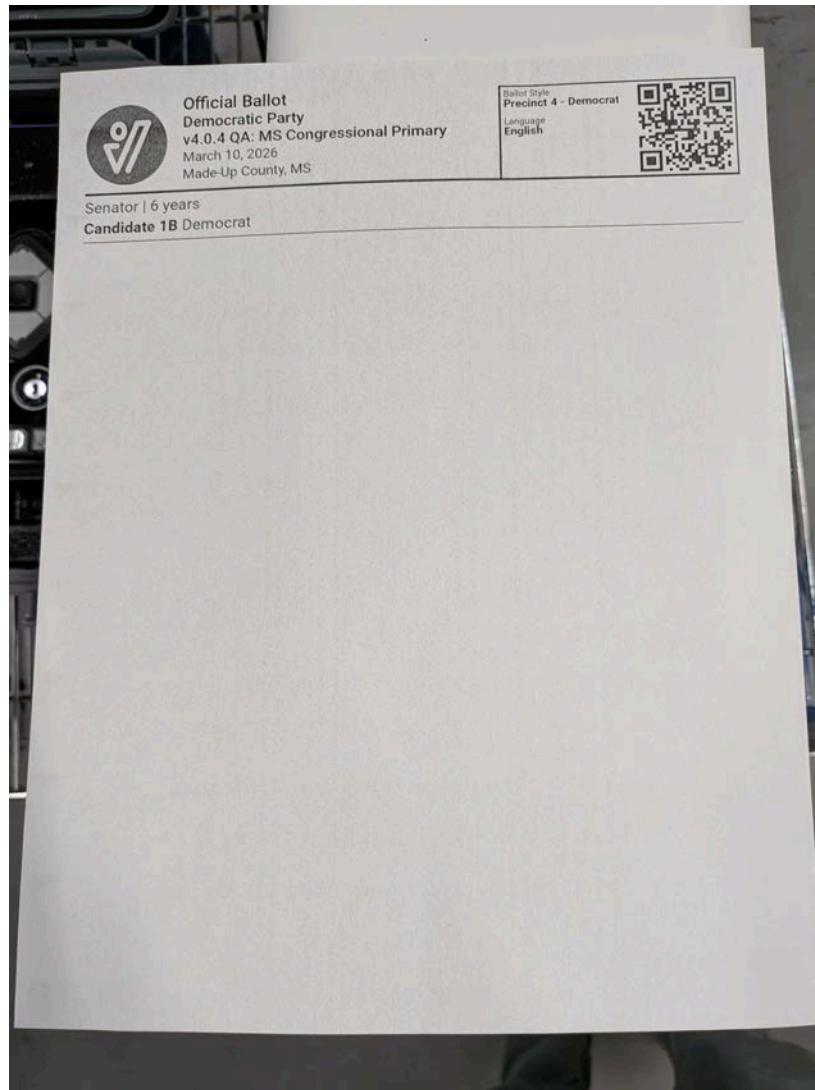
- b. Test conditions followed the intended set point (SP) for the process variables (PV) of temperature and with humidity. This is shown in the graph below of the chamber data log:

Temperature & %RH vs. Time, Cold Storage

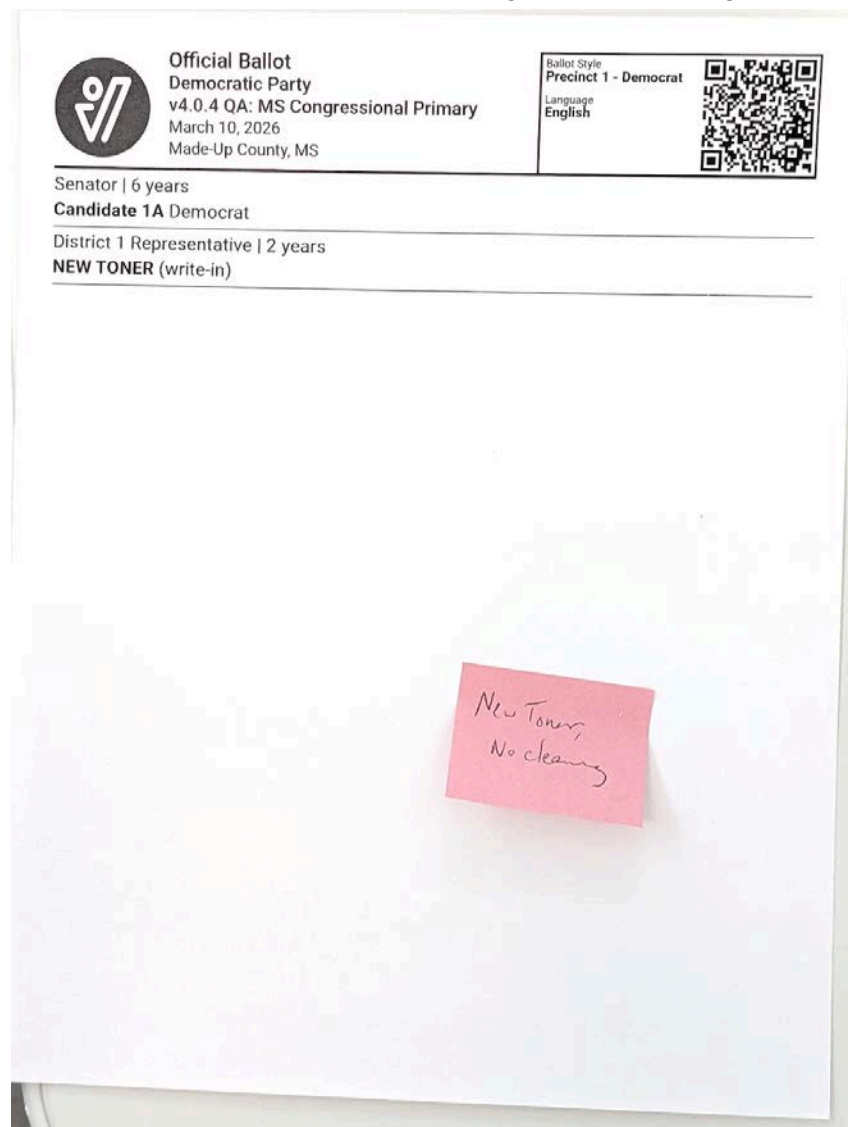


5. Normal function was confirmed for the UPS and printer after cold storage, if toner was replaced.

- a. See photos of the UPS and printer after hot storage in [Appendix A](#) and [Appendix B](#).
- b. After resting for several hours, the UPS and printer were tested with the VxMark BMD. Results are documented in spreadsheet [Appendix E](#).
- c. All essential UPS functions worked as expected.
- d. Printer functions worked mostly as expected, with the exception of a gray layer of toner or ink over the printable area on the first several prints after hot storage. This did not appear previously nor after cold tests. This printing artifact did not prevent the printed ballots from being properly scanned, interpreted, or tabulated in VxScan, but were still very visible and not proper behavior. Additional test prints several hours later showed that the printer continued to print with artifacts that potentially could cause concern. See image below for an example test print with a layer of gray over the page:



- e. Replacing the HP148A toner with one that was properly stored eliminated the artifacts, as shown in the print below right after replacing the toner:



Conclusions

- The Goldenmate 1500VA/1000W UPS is likely to withstand storage in extreme cold and hot temperatures as defined by VVSG2.0.
- The HP4001dn laser printer is likely to withstand storage in extreme cold and hot temperatures as defined by VVSG2.0, as long as the toner is replaced following hot storage in temperatures greater than 104°F (40°C). This threshold for replacing the

toner comes from the rated storage temperatures of the standard HP148A toner normally used with the HP4001dn printer.

- User instruction and documentation should clarify that standard maintenance procedures for the HP4001dn include replacing the toner when the toner (or printer holding the toner) is stored at temperatures greater than 104°F (40°C).

Appendix A: Photos of UPS

UPS Before Cold Storage









UPS After Cold Storage







UPS After Hot Storage







Appendix B: Photos of Printer

Printer Before Hot Storage







Printer After Hot Storage





Appendix C: VxMark Normal Function Checklist - UPS Cold storage - before 260427

Corresponds to detailed doc here:		<i>VxMark Tests of Normal Function</i>	
	Person(s) testing & location:	Pius Wong, Austin Reliability Labs, Austin	
	Date & time tested	4/27/2026	9:00 AM
	System tested & Unit/Model No.:	UPS: JUO2HE390707; Printer: VNL0728678; VxMark: MK-14-018	
	Before what other test?	Cold storage, 4-hour dwell, unpackage d	
	After what other test?	Thermals, drop test	
Always do these:			
No.	Test	Pass / Fail / Mixed?	Notes
1	Marks and prints valid ballots.	PASS	UPS functions normally on boot and print
2	Visually displays instructions and accurate feedback to users.	PASS	
3	Audibly plays instructions and accurate feedback to users.	PASS	
4	Accepts user inputs on the touchscreen.	PASS	
5	Accepts user inputs on the ATI and PAT.	PASS	
6	Accepts inputs on the barcode scanner.	PASS	
Sometimes do these (after environmental, vibrational, or other mechanical tests):			
No.	Test	Pass / Fail / Mixed?	Notes

7	Maintains physical security, restricting administrative access and preventing tampering without evidence.	PASS	
8	Protects internal electronics and subsystems.	PASS	
Inspection of parts, after tests of normal function above:			
No.	Part	Visual condition ?	notes
1	Computer	PASS	
2	Touchscreen	PASS	
3	Power Supply, internal	PASS	
4	Smart Card Reader	PASS	
5	USB-A receptacle	PASS	
6	USB 3 Switch	PASS	
7	Barcode Scanner	PASS	
8	Cable Assemblies	PASS	
9	Uninterruptible Power Supply (external to unit)	PASS	UPS functioning normally at 100% charge
10	Printer (external to unit)	PASS	normal print, small specks of ink/toner on left part of page, could clean later
11	Outer case	PASS	
12	Inner panels around electronics	PASS	
13	USB cover	PASS	
14	Security tie points	PASS	
15	Fasteners, both external and internal	PASS	
16	Wires, ports, and electrical connections	PASS	
18	other....		Generally looked inside after previous tests

Appendix D: VxMark Normal Function Checklist - UPS Cold storage - after 260428

Corresponds to detailed doc here:		<i>VxMark Tests of Normal Function</i>	
	Person(s) testing & location:	Pius Wong, Austin Reliability Labs, Austin	
	Date & time tested	4/28/2026	8:00 AM
	System tested & Unit/Model No.:	UPS: JUO2HE390707; Printer: VNL0728678; VxMark: MK-14-018	
	Before what other test?	Hot storage, 4-hour dwell, unpackaged	
	After what other test?	Cold storage, 4-hour dwell, unpackaged	
Always do these:			
No.	Test	Pass / Fail / Mixed?	Notes
1	Marks and prints valid ballots.	PASS	UPS functions normally on boot and print, after going back to ambient for 2 hours
2	Visually displays instructions and accurate feedback to users.	PASS	
3	Audibly plays instructions and accurate feedback to users.	PASS	
4	Accepts user inputs on the touchscreen.	PASS	
5	Accepts user inputs on the ATI and PAT.	PASS	
6	Accepts inputs on the barcode scanner.	PASS	
Sometimes do these (after environmental, vibrational, or other mechanical tests):			
No.	Test	Pass / Fail / Mixed?	Notes
7	Maintains physical security, restricting administrative access and preventing tampering without evidence.	PASS	

8	Protects internal electronics and subsystems.	PASS	
Inspection of parts, after tests of normal function above:			
No.	Part	Visual condition ?	notes
1	Computer	PASS	
2	Touchscreen	PASS	
3	Power Supply, internal	n/a	not opened up to inspect
4	Smart Card Reader	PASS	
5	USB-A receptacle	PASS	
6	USB 3 Switch	n/a	not opened up to inspect
7	Barcode Scanner	PASS	
8	Cable Assemblies	n/a	not opened up to inspect
9	Uninterruptible Power Supply (external to unit)	PASS	UPS functioning normally at 100% charge
10	Printer (external to unit)	PASS	normal print, small specks of ink/toner on left part of page, could clean later
11	Outer case	PASS	
12	Inner panels around electronics	n/a	not opened up to inspect
13	USB cover	PASS	
14	Security tie points	PASS	
15	Fasteners, both external and internal	PASS	not opened up to inspect
16	Wires, ports, and electrical connections	PASS	not opened up to inspect
18	other....		

Appendix E: VxMark Normal Function Checklist - UPS & Printer Hot storage - after 260428

Corresponds to detailed doc here:		<i>VxMark Tests of Normal Function</i>	
	Person(s) testing & location:	Pius Wong, Austin Reliability Labs, Austin	
	Date & time tested	4/28/2026	4:30 PM
	System tested & Unit/Model No.:	UPS: JUO2HE390707; Printer: VNL0728678; VxMark: MK-14-018	
	Before what other test?	general storage; continuous operations	
	After what other test?	Hot storage, 4-hour dwell, unpackaged	
Always do these:			
No.	Test	Pass / Fail / Mixed?	Notes
1	Marks and prints valid ballots.	PASS	UPS functions normally on boot and print, after going back to ambient for 2 hours
2	Visually displays instructions and accurate feedback to users.	PASS	
3	Audibly plays instructions and accurate feedback to users.	PASS	
4	Accepts user inputs on the touchscreen.	PASS	
5	Accepts user inputs on the ATI and PAT.	PASS	
6	Accepts inputs on the barcode scanner.	PASS	
Sometimes do these (after environmental, vibrational, or other mechanical tests):			
No.	Test	Pass / Fail / Mixed?	Notes
7	Maintains physical security, restricting administrative access and preventing tampering without evidence.	PASS	

8	Protects internal electronics and subsystems.	PASS	
Inspection of parts, after tests of normal function above:			
No.	Part	Visual condition ?	notes
1	Computer	PASS	
2	Touchscreen	PASS	
3	Power Supply, internal	n/a	not opened up to inspect
4	Smart Card Reader	PASS	
5	USB-A receptacle	PASS	
6	USB 3 Switch	n/a	not opened up to inspect
7	Barcode Scanner	PASS	
8	Cable Assemblies	n/a	not opened up to inspect
9	Uninterruptible Power Supply (external to unit)	PASS	UPS functioning normally at 100% charge
10	Printer (external to unit)	PASS	grayish layer of ink/toner over whole print; must replace toner even though ballot scanned
11	Outer case	PASS	
12	Inner panels around electronics	n/a	not opened up to inspect
13	USB cover	PASS	
14	Security tie points	PASS	
15	Fasteners, both external and internal	PASS	not opened up to inspect
16	Wires, ports, and electrical connections	PASS	not opened up to inspect
18	other....		